# Moving Home: Non-Market Insurance and Labour Market Risk*

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**Abstract**

The ability to move in with a parent provides insurance against labour market risk by allowing workers to reduce housing costs following income loss. Using linked panel data from the Health and Retirement Study, I show that job separations significantly increase the likelihood of moving home, and that this relationship persists into middle age even after accounting for parental health and caregiving needs. Motivated by this evidence, I estimate a structural life-cycle model in which individuals insure against unemployment risk through savings or by exercising the option to co-reside in a directed search environment. Access to co-residence enables workers to search for higher-wage jobs at the cost of longer expected unemployment durations; removing this option shortens search by roughly 5 weeks but lowers earnings and consumption. For an ex ante individual, the welfare value of this insurance is equivalent to a 71 percent increase in unemployment insurance benefits and remains stable from ages 20 to 40. Among younger workers, low asset holdings constrain self-insurance and make co-residence an important means of maintaining consumption during unemployment; among older workers, the reduced number of remaining working years limits their ability to rebuild earnings following job loss, increasing the value of the option to move home.

**Keywords:** co-residence, shared housing, labour market risk, non-market insurance

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# 1 Introduction

Households of at least two adult generations are common in the U.S. economy, with 18 percent of households in 2021 including at least two adult generations (Cohn et al., 2022). One reason for these arrangements could be that co-residence provides informal insurance against labour market risk by lowering fixed housing costs during periods of income loss. This mechanism is particularly relevant in an expensive housing environment: in 2023, more than half of renter households spent over 30 percent of their income on housing (U.S. Census Bureau, 2024). High housing costs amplify the welfare consequences of income shocks such as unemployment, increasing the value of informal insurance arrangements like co-residence. This paper quantifies how the value of this insurance mechanism varies across the life cycle and its impact on consumption, savings, and job search decisions.

I document this housing-based insurance mechanism by examining how transitions into unemployment raise the likelihood of co-residence with parents, and show that this relationship persists into middle age (up to age 40). Workers are roughly twice as likely to move home when they transition from employment to unemployment than when they remain employed, a pattern that remains robust after controlling for parent and child characteristics and persists well into middle age, when co-residence motives are harder to disentangle. Motivated by this finding, I estimate a structural model of job search and family co-residence to identify the effects of this option on search behaviour and welfare. The model shows that access to co-residence leads workers to search in submarkets offering higher wages with longer expected search durations. For an ex ante consumer, the option to move home is equivalent in welfare terms to approximately a seventy percent increase in unemployment insurance benefits.

Existing work from Kaplan (2012) identifies this insurance channel among young men (ages 17–22). A major challenge in extending this understanding to older workers is disentangling the motives for moving home. While existing empirical work documents co-residence as an insurance mechanism for young adults (Acolin et al., 2024; Albanesi et al., 2022; García-Andrés et al., 2021),

the motives of older children are less clear. Shared housing acts as an in-kind transfer from parents to their adult children by reducing housing costs and diminishing the impact of negative income shocks. However, co-residence can also represent an in-kind transfer from children to their parents if they provide household labour and especially eldercare that the parent would otherwise need to purchase on the market. The degree to which parental needs are a confounding factor for identifying the labour market insurance mechanism increases with the ages of parents and children. Applying this insurance channel to workers into middle age therefore requires controlling for parental health and caregiving needs, which I address using detailed information from the Health and Retirement Study.

The Health and Retirement Study (HRS) offers an exceptional foundation for examining shared housing as a response to labour-market risk. Its primary advantage lies in the long-term linkage of parents and their adult children, a feature preserved even when children establish independent households. These linkages are maintained for roughly two decades with a consistent set of observables, enabling the study of co-residence, transfers, and parental characteristics over an extended time horizon.[1] Although the HRS is oriented toward health and aging rather than earnings dynamics or unemployment, its sustained intergenerational design makes it uniquely well suited for studying co-residence as an informal insurance mechanism.

An additional strength of the HRS is its rich measurement of parental health and functional limitations, which is essential for separating labour market insurance from caregiving-related co-residence. Among older adult children, living with a parent may reflect responses to parental health shocks or emerging eldercare needs rather than insurance against their own income risk. The HRS provides diagnoses, health histories, limitations to independent living, and detailed reports of children's realized and anticipated caregiving, allowing me to control directly for these confounding motives.

[1] A natural alternative is the National Longitudinal Survey of Youth 1997 cohort, which also follows parents and children over a similar period. Yet as the NLSY97 cohort ages, many variables central to co-residence and intergenerational support are discontinued, limiting its usefulness for analyses focused on older adults. Among young adults, the NLSY97 confirms this paper's main empirical observation: children experiencing a job separation are far more likely to move home than their peers who remain employed.

This feature also motivates restricting the model's insurance option to ages below 40, when in the data parental health needs are less salient and co-residence remains more clearly interpretable as a labour market insurance mechanism. By conditioning on these health-related factors, I isolate the insurance motive beyond only young adults.

I use these empirical observations to motivate a quantitative model of piece-rate[2] job search and co-residence. In this environment, consumers vary in their individual-specific productivity and asset accumulation, which is tied to a savings choice. While unemployed, working age individuals choose a submarket in which to search every period. These submarkets are identified on three dimensions: the worker's age, which is observable by the firm and relevant for expectations of future match output; the worker's productivity; and the piece-rate of match output. Additionally, workers choose whether to live at home with a parent or independently until age 40. After that time, the option to move home is withdrawn, motivated by the empirical patterns discussed above. When able, children trade off some utility benefit of living independently against a financial cost of purchasing housing services on the private market. They additionally trade off wages and unemployment risk via their search choice; submarkets with higher job-finding probabilities offer lesser piece-rates and vice versa.

Parameters estimated internally to the model target moments from the U.S. Bureau of Labor Statistics, U.S. Bureau of Economic Analysis, and earnings distribution estimates from Kuhn, Ríos-Rull, et al., 2016. The utility benefit of independence is calibrated to target the aggregate share of workers in the Health and Retirement Study who move home following a job separation. The calibrated model can generate untargeted age-dependent co-residence flows that increase during employment-to-unemployment transitions from ages 20 to 40.

When consumers have the option to move home, they select into submarkets with higher expected wages but longer search durations, consistent with co-residence providing insurance against labour market risk. This insurance allows workers to tolerate longer periods of unemployment

[2]In search and matching theory, the piece-rate of a match between a worker and a firm is the share of the match output that is earned as wages by the worker.

in exchange for higher-paying matches. Removing this option shortens average search duration by approximately five weeks and increases job-finding and employment rates. However, these improvements in employment outcomes come at the cost of lower accepted wages on average. As a result, despite higher employment, equilibrium mean consumption and asset holdings decline.

I evaluate the welfare implications of the co-residence option by comparing the calibrated baseline economy to one in which all workers must live independently and pay a fixed housing cost. The welfare value of this option is largest when unemployment insurance (UI) benefits are low, indicating that co-residence partially fills the insurance gap left by limited UI. In aggregate, access to co-residence is equivalent, in ex ante welfare terms, to a 71 percent increase in UI benefits. This value persists over the life-cycle: workers aged 36–40 value the option at the equivalent of a 79 percent increase in UI benefits. This pattern reflects the greater vulnerability of older workers to income shocks, as they have fewer remaining working years to recover from unemployment and face a predictable decline in income at retirement. In the model, this mechanism operates through older workers' reduced reliance on self-insurance via saving following an employment-to-unemployment transition, which increases the value of alternative insurance channels such as co-residence.

**Related Literature** Existing work in quantitative macroeconomics tends to take one of two approaches when modeling the co-residence choice among adult children. The first approach is to construct a strategic game between parents and children, and characterize the effects of economic characteristics on the sequential choices of both agents. Models of this kind can be considered as evaluating behaviour under a joint co-residence choice in static and dynamic settings (Kaplan, 2012, Zaber, 2018, Albanesi et al., 2022). In contrast, the second approach abstracts from strategic interaction between parents and children and instead evaluates the existence of the co-residence option as it affects other macroeconomic outcomes, typically in a dynamic setting (Dyrda et al., 2024). This paper takes the second approach, extending this literature by endogenizing labour supply decisions over the life-cycle in a directed search environment. Further, I estimate the value associated with the option to move home in the context of existing unemployment insurance

programs.

The remainder of this paper proceeds as follows: Section 2 provides an overview and comparison of both the HRS and NLSY97, as well as identification of the labour market co-residence motive. Section 3 outlines the theoretical model of job search and non-market co-residence and section 4 explains the model estimation. Section 5 discusses the model validation and results; section 6 concludes.

# 2 Empirical Motivation

A primary challenge in identifying co-residence as a form of non-market insurance is separating returns to the parental home prompted by labour market shocks from co-residence driven by child-care, divorce, or eldercare. From the perspective of the adult child, moving home may follow a divorce or separation, particularly when outside housing options are costly. Among aging parents, co-residence may arise in response to increasing health or caregiving needs. Addressing this requires data that link parents and adult children over time and provides sufficient information on parental health and caregiving needs to rule out confounding motives. This section describes the data, establishes the empirical relationship between unemployment and co-residence, and clarifies the identification challenges it raises.

## 2.1 Data

Identifying the labour market motives for co-residence requires data that simultaneously track adult children and aging parents over long horizons, making the Health and Retirement Study (HRS) uniquely suited to this analysis. The HRS is a biennial panel of roughly 20,000 U.S. households beginning in 1998, and its design provides repeated linkages between parents and adult children as both age. Crucially, the survey also collects unusually rich information on parental health, functional limitations, and caregiving requirements. This combination of intergenerational linkages and detailed aging-related measures is essential for distinguishing co-residence driven by labour

market motivations versus those motivated by eldercare or parental need.

I use the RAND HRS Family File, which links the sample of parent households to 55,000 children at the individual level through 2018. Co-residence and employment transitions are constructed using adjacent survey waves. Because the HRS interviews every two years, I drop the initial 1998 wave to ensure all transitions are observed over two-wave intervals.

### 2.2 Sample Selection

The sample restrictions are designed to isolate co-residence decisions that reflect the child's labour market conditions rather than parental need. The likelihood that co-residence reflects eldercare increases with the ages of both parents and children. To minimize this confounding channel, the sample is restricted to children aged 20–40, and post-secondary students are excluded. This age range preserves the insurance mechanism of interest while removing most cases in which parents have substantial caregiving needs.

Table 1 summarizes characteristics of children aged 20–40 across employment transitions. Demographic differences between employed-to-unemployed (EU) and employed-to-employed (EE) workers are modest, but the share of children who move home is substantially higher among those experiencing an EU transition. Before conditioning on any covariates, children who separate from employment move home at nearly three times the rate of those who remain employed. EU children are also more likely to receive financial transfers from parents, consistent with parents providing support during periods of income loss.

Eldercare is not a major confounder in this age range. Assistance with activities of daily living (ADLs) and instrumental activities of daily living (IADLs) is rare, typically 1–2 percent, and is slightly more common among children who remain employed. These patterns indicate that co-residence among 20–40-year-olds primarily reflects the child's labour market circumstances rather than parental health or caregiving needs.

In the first two panels of demographics and co-residence, there is relatively little variation in

Table 1: Summary Statistics for Children Aged 20–40 in the HRS

| | All Children | EU Type | EE Type |
|---|---|---|---|
| **Panel A: Demographics** | | | |
| Age | 32.1 | 31.9 | 33.2 |
| Years of Education | 13.6 | 13.0 | 12.2 |
| **Panel B: Co-residence** | | | |
| Coresidence Share | .111 | .155 | .184 |
| Move Home Share | .018 | .057 | .020 |
| **Panel C: Financial Transfers** | | | |
| Share who Give to Parents | .018 | .020 | .015 |
| Share who Receive from Parents | .215 | .233 | .163 |
| Mean Transfer from Parents | $1905.44 | $1671.21 | $1292.77 |
| **Panel D: Eldercare** | | | |
| Share who Help with ADLs | .006 | .010 | .015 |
| Share who Help with IADLs | .009 | .015 | .021 |
| Share who Intend to Help in Future | .329 | .343 | .345 |

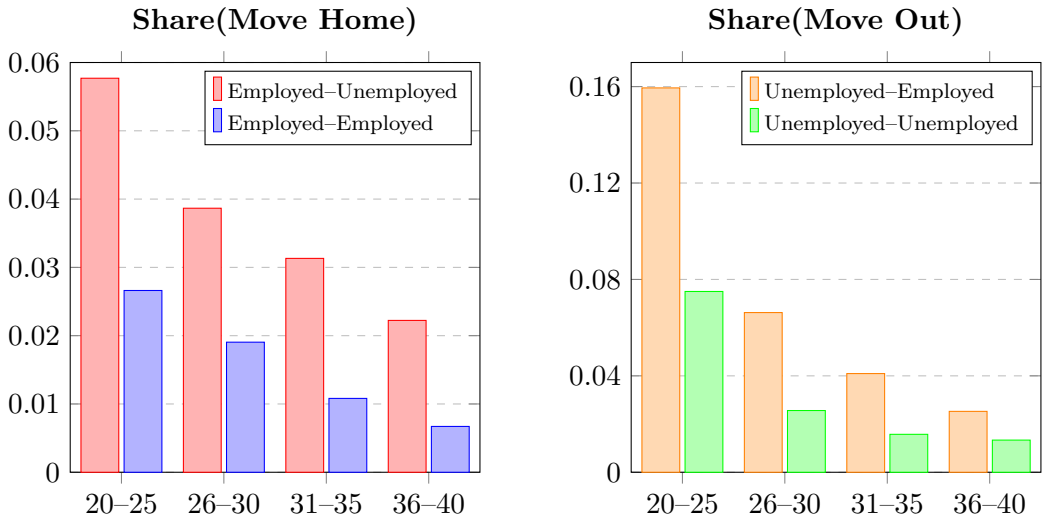
**Note:** Panel A reports mean child age and years of education. Panel B shows the static co-residence share and share of children who move home. Panel C lists the share of children who give and receive a transfer from their parent(s), as well as the size of the average transfer amount from a parent to their child. Panel D summarizes the share of children who assist with activities of daily living, instrumental activities of daily living, and their intention to help with eldercare in the future.

most characteristics between EU, EE, and all children. However, the share of children who move home is much larger among EU children, suggestive of the key empirical observation summarized in Figure 1. Children transitioning out of employment move home at a rate 2.9 times larger than those who remain employed, before controlling for selection or further confounding characteristics.

One feature of the HRS that is useful when evaluating parental support is direct financial transfers between parents and children. This type of direct support also interacts with employment transitions since it functions as an insurance mechanism similarly to co-residence. EU children are 43 percent more likely to receive a transfer from a parent and on average receive around $400 more than EE peers.

The final panel in Table 1 is informative of parent needs driving co-residence. It reports child assistance with activities of daily living and instrumental activities of daily living, as well as fu-

Figure 1: Share Who Move Home and Move Out by Age and Employment Transition




**Note:** Flows into and out of the parental household among job-losers (EU) and job-finders (UE). For each age group, consumers transitioning out of employment move home at greater rates than their employed peers. Similarly, consumers transitioning into employment move out at greater rates than those who remain unemployed. In addition, both types of co-residence flows decline as adult children age.

ture intention to help. The former two are measures used by the National Institute of Health to summarize necessary capabilities for aging individuals to live independently. Activities of daily living refer to the most basic activities including feeding oneself, mobility within the household, and personal hygiene. Instrumental activities refer to more complex tasks such as transportation, managing finances, and meal preparation. Panel D shows there is little overlap between EU transitions and additional eldercare for parents: EE children are helping aging parents at higher rates than EU. The relative size of these shares is also small, peaking at around 2 percent of children aiding parents. This indicates that the age restriction on the sample of children filters out most eldercare relationships while preserving co-residence as labour market insurance for children.

### 2.3 Labour Market Co-Residence Motive

The empirical contribution of this paper is centered around the observation that a larger share of adult children move home when experiencing a transition from employment to unemployment

relative to peers who are persistently employed. This trend, shown in Figure 1, is representative of non-market insurance against labour market risk via in-kind housing services. Existing work from Kaplan (2012) identifies this mechanism for non-college men aged 17 to 22. I expand this understanding to all workers up to age 40 while controlling for parent and child characteristics.

Table 2: Probit Predicting Move Home Likelihood

| **Independent Variable** | **AME** | **p-value** |
|---|---|---|
| **Child Employment Transition** | | |
| EE | *Base* | |
| EU | 0.016 | 0.000 |
| UE | 0.004 | 0.040 |
| UU | 0.003 | 0.079 |
| **Child Income** | | |
| <10K | *Base* | |
| 10–35K | −0.012 | 0.000 |
| 35–70K | −0.020 | 0.000 |
| 70–100K | −0.020 | 0.000 |
| 100K+ | −0.024 | 0.000 |
| **Child Age** | | |
| 20–25 | *Base* | |
| 26–30 | −0.006 | 0.001 |
| 31–35 | −0.008 | 0.000 |
| 36–40 | −0.012 | 0.000 |
| Parent Assets | $-1.84 \times 10^{-9}$ | 0.021 |
| Parent Home Value | $3.86 \times 10^{-9}$ | 0.002 |
| **Parent Employment** | | |
| Not Working | *Base* | |
| Part-time Work | 0.002 | 0.457 |
| Full-time Work | 0.003 | 0.008 |

**Note:** Coefficients are expressed as probability-unit average marginal effects. For the full regression specification and table of covariates, see Appendix A.1.

To identify the existence of this labour market insurance channel via non-market housing arrangements, I employ a probit regression of cross-wave employment flows on workers' move home choice, shown in Table 2. A worker is considered to have "moved home" if in one survey period they

are living independently and in a subsequent period are living with a parent. The primary covariates are employment flows: specifically for workers who are employed through two subsequent periods (EE) and those who separate from employment for any reason (EU). Additional controls include child characteristics such as income, education, marital status, and financial transfers from parents as well parent characteristics including assets, home values, education, and health conditions.

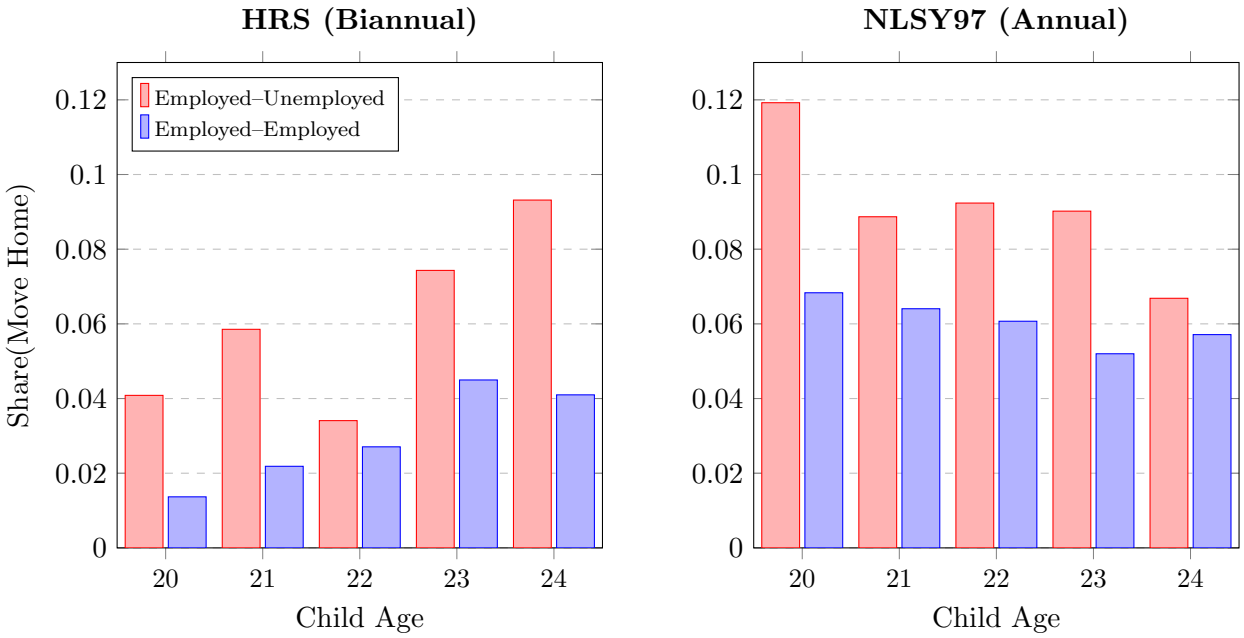
After controlling for these characteristics, workers experiencing an EU transition are 1.6 percentage points more likely to move home with a parent relative to an identical worker with an EE transition. The likelihood of an adult child moving home is decreasing in their income and age, suggesting that workers in an EU transition will self-insure via residual earnings from employment before they elect to move in with a parent. Children in the data also move out when they find a job. Specifically, workers in an Unemployed-Employed transition move out of co-residence at much higher rates than those who remain unemployed.

### 2.4 Additional Evidence from the NLSY97

Although the HRS is uniquely suited for identifying co-residence motives into middle age, it is less ideal for measuring short unemployment spells or search duration. To complement the HRS, I incorporate evidence from the National Longitudinal Survey of Youth 1997 Cohort (NLSY97), which provides high-frequency labour market histories and reports co-residence through 2009, when the oldest respondents were 29. The NLSY97 validates the main HRS finding: EU workers are substantially more likely to move home, as shown in Figure 2. Additionally, the data include measures of unemployment spell length: living at home is associated with roughly 1.4 additional weeks of job search, consistent with reduced liquidity constraints and more selective search.

The NLSY97 reports the length of unemployment spells for youth moving between employers, which can be interpreted as a measure of search duration. Since employment and search are very noisy in the HRS, I leverage this feature of the NLSY97 to characterize the effects of co-residence on search behaviour. Intuitively, one can expect that children living at home are less financially

Figure 2: Comparison of Move Home Trends Among the HRS and NLSY97




**Note:** At each age from 20 to 24, workers experiencing a job separation (EU) are more likely to move home relative to their peers who remain employed (EE). Additionally, there is a decreasing trend in move home flows by age present in the NLSY97. This pattern is also present in the HRS, but over a longer range of ages as shown in the first panel of Figure 1. For further discussion of the comparability of the HRS and NLSY97, see appendix A.2.

constrained due to shared or avoided housing costs. This reduction of expenses affects the search process in two ways: first, workers may search less intensely, resulting in longer search times. Second, workers may reject otherwise viable offers if they are below their expected quality, both extending the search process but increasing wages conditional on a match.

I estimate a simple ordinary least-squares regression of co-residence status on the average search duration in weeks. Coefficient estimates are shown in Table 3 and the most relevant is the marginal effect of co-residing with parents on search duration. After controlling for previous wages, age, education, and additional covariates, living at home is associated with an additional 0.76 weeks of search time before matching. The sample of workers is restricted to only those who have experienced an EU transition to account for recent job separation. Education and sex are significant predictors

Table 3: OLS Estimation of Co-Residence and Search Duration Among EU Workers

| **Independent Variable** | **Coefficient** | **p-value** |
|---|---|---|
| Co-Residence | 0.762 | 0.016 |
| **Age** | | |
| 20 | 0.667 | 0.683 |
| 21 | 0.762 | 0.640 |
| 22 | 1.179 | 0.480 |
| 23 | 1.549 | 0.360 |
| 24 | 2.302 | 0.180 |
| 25 | 3.476 | 0.049 |
| **Education** | | |
| Less Than High School | *Base* | |
| HS Graduate | −0.681 | 0.134 |
| Some College | −0.987 | 0.238 |
| Post-Secondary and Above | −1.149 | 0.043 |
| Gender (Female) | −0.696 | 0.008 |
| Enrolled in School | −1.834 | 0.000 |

**Note:** Living at home is associated with an additional 0.76 weeks of job search before finding a match. Coefficients on age reflect the relative change in search duration compared to a worker aged 18. For the full specification and results including year fixed effects, see Appendix A.3.

of search duration, but age is not. These results suggest that transitions around employment states are a key determinant of co-residence flows, and for this reason the quantitative model will embed a directed search framework where workers can influence both their wages and unemployment risk.

# 3 Model

This paper develops a life-cycle model of consumption, savings, and co-residence that extends the study of shared housing to older adult children and embeds the co-residence decision within a formal directed search framework. The key extension is to treat the option to move home as an age-dependent insurance channel that interacts with workers' job search behaviour and unemployment risk. To isolate this mechanism, the model abstracts from parental decision-making, thereby

removing strategic interactions through transfers or co-residence availability that feature in some existing parent–child models. While such richer frameworks capture a broader set of intergenerational dynamics, they are quantitatively demanding and difficult to apply over long horizons in a life-cycle setting. The streamlined structure adopted here allows for a tractable evaluation of how co-residence shapes job search, match quality, and welfare over the working life.

### 3.1 Consumers

Consumers are the primary decision-makers in this model and choose allocations for consumption ($c$) and savings ($a'$) at every age. While unemployed, they choose a submarket in which to search indexed by piece-rate ($\phi$), their age, and their individual-specific productivity ($\epsilon$). If employed, they earn an after-tax piece-rate of output as their wage, subject to an age component of productivity ($\nu_j$). If unemployed, consumers receive a government-financed unemployment benefit $b$; during retirement, they receive a similarly funded pension benefit $S$.

All consumers in this environment are considered to be adult children in the context of multigenerational co-residence. This means that for each consumer there is some unmodelled agent, a parent, who is older than the child and may provide non-market housing while they are able. When choosing where to live, consumers face a trade-off between independence utility ($\chi$) and the cost of rental housing ($\kappa_h$). If they live at home ($d_h = 0$), they can avoid the cost of housing but forgo the utility benefits of living independently ($d_h = 1$). This decision is subject to a Type-I extreme value taste shock ($\xi^c, \xi^i$).

$$U(a, \epsilon, \phi, a', d_h) = \frac{c^{1-\sigma}}{1-\sigma} + d_h\chi \tag{1}$$

subject to

$$c + a' = (1-\tau)\phi\epsilon\nu_j + (1+r)a - d_h\kappa_h$$

when employed, and

$$c + a' = b + (1 + r)a - d_h\kappa_h$$

when unemployed.

The problem for a worker can then be split into three cases: employed, unemployed, and searching for work. Unemployed workers search for a match first, followed by choices conditional on their employment state. When searching, workers choose piece-rate which along with their age and individual-specific productivity determines a submarket $(j, \epsilon, \phi)$. The value of search is a comprised of the values of employment and unemployment, weighted by the job-finding probability $f(j, \epsilon, \phi)$.

$$V_s(j, a, \epsilon) = \max_{\phi} \left\{ f(\theta(j, \epsilon, \phi))V_e(j, a, \epsilon, \phi) + [1 - f(\theta(j, \epsilon, \phi))]V_u(j, a, \epsilon) \right\} \tag{2}$$

Whether employed or unemployed, the consumer chooses savings and whether or not to live at home until age 40. After age 40, there is no option to live with parents. Those under age 40 face extreme value Type-I shocks on each option, co-residence and living independently, drawn from a common distribution. The value of unemployment prior to the realization of the shock is:

$$V_u(j, a, \epsilon) = \max \mathbf{E}_{\xi^c, \xi^i} \left\{ V_u^{cores}(j, a, \epsilon) + \xi^c, V_u^{indep}(j, a, \epsilon) + \xi^i \right\} \tag{3}$$

The probability of a consumer choosing to live independently $(h_u)$ will depend on the relative values of $V_u^{cores}$ and $V_u^{indep}$ as well as the EV scale parameter $\eta$ which determines the variance of the shock process.

$$Pr(d_h = 1|u) = h_u(j, a, \epsilon) = \frac{\exp\left(\eta V_u^{indep}(j, a, \epsilon)\right)}{\exp\left(\eta V_u^{indep}(j, a, \epsilon)\right) + \exp\left(\eta V_u^{cores}(j, a, \epsilon)\right)} \tag{4}$$

**I. Unemployed Problem**

Following the realization of the shocks, the unemployed consumer chooses saving for the current period before reentering the search process next period. This choice is summarized by a value function for each housing choice: co-residence and living independently. For an unemployed consumer who lives at home:

$$V_u^{cores}(j,a,\epsilon) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \beta V_s(j+1, a', \epsilon) \right\} \tag{5}$$

subject to

$$c + a' = b + (1+r)a$$

When the unemployed worker lives independently, their problem is summarized as:

$$V_u^{indep}(j,a,\epsilon) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \chi + \beta V_s(j+1, a', \epsilon) \right\} \tag{6}$$

subject to

$$c + a' = b + (1+r)a - \kappa_h$$

**II. Employed Problem**

The problem for an employed consumer is defined similarly. The consumer makes a discrete choice between co-residence and living on their own subject to extreme value shocks and their own age. The probability of such a consumer living independently is:

$$Pr(d_h = 1|e) = h_e(j,a,\epsilon,\phi) = \frac{\exp\left(\eta V_e^{indep}(j,a,\epsilon,\phi)\right)}{\exp\left(\eta V_e^{indep}(j,a,\epsilon,\phi)\right) + \exp\left(\eta V_e^{cores}(j,a,\epsilon,\phi)\right)} \tag{7}$$

where the discrete choice is given by:

$$V_e(j,a,\epsilon,\phi) = \max \mathbf{E}_{\xi^c,\xi^i}\left\{ V_e^{cores}(j,a,\epsilon,\phi) + \xi^c, V_e^{indep}(j,a,\epsilon,\phi) + \xi^i \right\} \tag{8}$$

After observing the co-residence shocks, the employed worker chooses their savings before facing

a job destruction shock $\delta$. If they are subject to the shock, they become unemployed and must search again in the following period. If they instead stay matched, the enter into employment next period without engaging in the search process. When an employed worker lives at home their value is:

$$V_e^{cores}(j,a,\epsilon,\phi) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \beta[(1-\delta)V_e(j+1,a',\epsilon,\phi) + \delta V_u(j+1,a',\epsilon)] \right\} \tag{9}$$

subject to

$$c + a' = (1-\tau)\phi\epsilon\nu_j + (1+r)a$$

As before, when living independently the employed worker's problem becomes:

$$V_e^{indep}(j,a,\epsilon,\phi) = \max_{a' \geq 0} \left\{ \frac{c^{1-\sigma}}{1-\sigma} + \chi + \beta[(1-\delta)V_e(j+1,a',\epsilon,\phi) + \delta V_u(j+1,a',\epsilon)] \right\} \tag{10}$$

subject to

$$c + a' = (1-\tau)\phi\epsilon\nu_j + (1+r)a - \kappa_h$$

**III. Retirement Problem**

When consumers reach age 65, they enter retirement and receive a constant pension benefit $S$. Each period they make a savings decision and face an age-dependent mortality risk $\psi_j$. The problem of retirees can be expressed as:

$$V_r(j,a) = \max_{a' \geq 0} \left\{ \frac{[S + (1+r)a - a']^{1-\sigma}}{1-\sigma} + \beta\psi_j V_r(j+1,a') \right\} \tag{11}$$

## 3.2 Firms

Firms choose whether or not to post in each submarket at a common cost $\kappa_p$. Submarkets are identified by workers' individual-specific productivity and a piece-rate of the match output offered

by firms. Additionally, the firm takes the worker's age into account when making the posting decision as it influences expectations of future output as well as an age component of individual-specific productivity. Once a match is formed, it persists until a job destruction shock is received with probability $\delta$ or the worker reaches retirement age. The firm's value for a match can be expressed as:

$$V_f(j,\epsilon,\phi) = \epsilon\nu_j(1-\phi) + \beta[(1-\delta)V_f(j+1,\epsilon,\phi) + \delta V_p(j+1,\epsilon,\phi)] \tag{12}$$

where $q(\theta(j,\epsilon,\phi))$ is the firm's job filling probability and $\theta = \frac{v}{u}$ is a measure of market tightness. Matching in this environment is determined via a Den Haan et al. (2000) function of the form:

$$M(u,v) = \frac{uv}{(u^\alpha + v^\alpha)^{1/\alpha}}$$

where job filling and finding probabilities are:

$$q(\theta(j,\epsilon,\phi)) = \frac{M(u,v)}{v} = \left[\frac{1}{1+\theta^\alpha}\right]^{1/\alpha}$$
$$f(\theta(j,\epsilon,\phi)) = \frac{M(u,v)}{u} = \left[\frac{1}{1+(1/\theta)^\alpha}\right]^{1/\alpha}$$

A submarket will have vacancies posted as long as there are non-negative expected profits from operating. In other words, the firm must produce enough to cover the cost of posting. This entry decision can be expressed as:

$$V_p(j,\epsilon,\phi) = \max\{q(\theta(j,\epsilon,\phi))V_f(j,\epsilon,\phi) - \kappa_p, 0\} \tag{13}$$

In equilibrium, expected profits will be zero and $V_p(j,\epsilon,\phi) = 0$ for all $(j,\epsilon,\phi)$ submarkets where the firm posts.

### 3.3 Equilibrium

**Definition 1.** Given an exogenous interest rate $r$ and a tax rate $\tau$, an equilibrium in this model is characterized by policy functions $c_u(s_u), c_e(s_e), c_r(s_r), h_u(s_u), h_e(s_e),\ a'_u(s_u), a'_e(s_e), a'_r(s_r), \phi(s_u)$ and stationary distributions of consumers $\Omega_u(s_u), \Omega_e(s_e), \Omega_r(s_r)$ such that:

1. The policy function $\phi(s_u)$ solves the problem of the searcher given in (2)
2. The policy functions $c_u(s_u), a'_u(s_u), h_u(s_u)$ solve the problem of the unemployed consumer given in (3)
3. The policy functions $c_e(s_e), a'_e(s_e), h_e(s_e)$ solve the problem of the employed consumer given in (8)
4. The policy functions $c_r(s_r), a'_r(s_r)$ solve the problem of the retired consumer given in (11)
5. The zero-profit condition (13) must hold in all submarkets where firms post vacancies
6. The government sets a proportional tax on labour income ($\tau$) which satisfies their budget constraint

$$\int \tau\phi\epsilon\nu_j \; d\Omega_e(j, a, \epsilon, \phi) = \int b \; d\Omega_u(j, a, \epsilon) + \int S \; d\Omega_r(j, a) \tag{14}$$

## 4 Calibration

The main externally estimated parameter in this model is the job destruction rate ($\delta$), which is computed from the annualized share of individuals who transition out of employment. Consumer risk aversion ($\sigma$), discount factor ($\beta$), and interest rate ($r$) are taken at values standard in the literature; match elasticity ($\alpha$) is imported from the seminal paper by Den Haan et al. (2000). Series for survival probabilities are taken from the Social Security Administration's Bell and Miller (2005) life tables and the age profile of earnings is estimated in Raveendranathan and Stefanidis (2025) using the Panel Study of Income Dynamics.

Table 4: External Parameters

| Parameter | | Value | Source |
|---|---|---|---|
| **Panel A: Housing** | | | |
| $\kappa_h$ | Rent to income ratio | 0.05 | |
| $\eta$ | EV distribution scale | 1.000 | |
| **Panel B: Labour Market** | | | |
| $\alpha$ | Match elasticity | 1.27 | Den Haan et al. (2000) |
| $\delta$ | Job destruction rate | 0.02 | EU transition share (HRS) |
| $\nu_j$ | Age profile of earnings | | Panel Study of Income Dynamics |
| **Panel C: life-cycle** | | | |
| $\sigma$ | Risk aversion | 2.00 | |
| $\beta$ | Discount factor | 0.96 | |
| $r$ | Interest rate | 0.04 | |
| $\psi_j$ | Survival probabilities | | Social Security Administration |

Table 5: Internal Parameters

| Parameter | | Value | Target | Model | Data |
|---|---|---|---|---|---|
| $\chi$ | Utility of independence | 4.703 | EU move home share | *.0413* | *.0413* |
| $\kappa_p$ | Vacancy posting cost | .0034 | Unemployment share[a] | *.0478* | *.0420* |
| $b$ | UI benefit | .2454 | UI expenditure-to-income ratio[b] | *.0042* | *.0042* |
| $S$ | Social security benefit | .7537 | SS expenditure-to-income ratio[c] | *.0525* | *.0525* |
| $\sigma_\epsilon$ | Std. dev. productivity | .8868 | Std. dev. of log earnings (age 26–30)[d] | *.9000* | *.9000* |

[a] U.S. Bureau of Labor Statistics (2025)
[b] U.S. Bureau of Economic Analysis (2024c)
[c] U.S. Bureau of Economic Analysis (2024c)
[d] Kuhn, Ríos-Rull, et al. (2016)

A summary of the internal parameter estimates can be found in Table 5. The most important estimate from the perspective of the co-residence-based insurance is the independence utility, $\chi$, which is targeted to match the share of EU workers who move home in the HRS. The cost of posting, $\kappa_p$, is calibrated to target the unemployment share and the standard deviation of productivity, $\sigma_\epsilon$, is calibrated to target the standard deviation of log earnings for workers aged 26 to 30. The fiscal system is estimated via unemployment insurance ($b$) and pension benefits ($S$), which are targeted to aggregate spending on UI and pensions, respectively.

# 5 Results

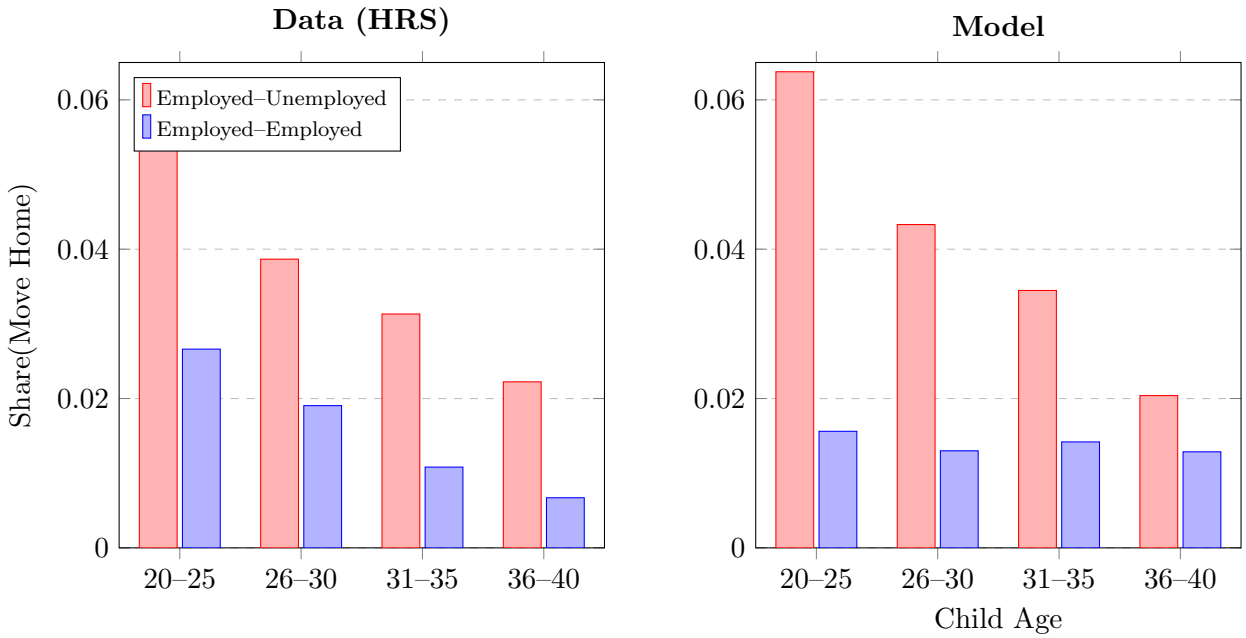
This section evaluates joint flows of employment and co-residence generated by the quantitative model. These flows are compared against the empirical trend of EU workers moving home in greater proportions than EE. Further, it outlines positive and normative results concerning job search behaviour, model aggregates, welfare in terms of unemployment insurance benefits, and heterogeneity in welfare by age.

## 5.1 Model Validation

The quantitative model can reproduce two empirical patterns that provide evidence for housing decisions acting as a form of labour market insurance. To show this, I compute the share of consumers who move home while also experiencing an employment to employment (EE) transition relative to those who experience an employment to unemployment transition (EU). These simulated move home flows are contrasted against comparable moments from the HRS in Figure 3.

Across age groups, EU consumers are significantly more likely to move home than their EE peers. This is indicative of moving home acting as an insurance mechanism against these labour market transitions and associated earnings risk. In an environment where housing costs represent a significant share of consumption during unemployment, moving home can be optimal even when forgoing independence utility.

Figure 3: Comparison of Move Home Flows Between Model and Data




**Note**: Move home flows are larger for EU consumers and are decreasing in age.

The magnitude of this effect is also decreasing in the consumer's age for the EU type; this has several implications. As consumers age, they are more able to self-insure in their household through saving rather than co-residing. If living independently is sufficiently rewarding, EU workers will dissave before they move home. Another reason age influences the move home choice among workers is because it interacts with incomes via individual-specific productivity. As workers age, they become richer as their productivity rises.

## 5.2 Job Search Behaviour and Model Aggregates

To quantify the role of co-residence in shaping workers' behaviour, I use the model to construct a counterfactual economy in which no individual can move home with their parents at any age. I then compare aggregate outcomes in this counterfactual to those in the calibrated baseline. Removing the co-residence option has meaningful consequences for search behaviour, asset accumulation, and

consumption smoothing. Without access to parental housing, unemployment shocks translate more directly into reductions in non-housing consumption, as workers must rely solely on their own assets to buffer income losses. In response to this tighter liquidity environment, job seekers reallocate toward submarkets with higher job-finding probabilities and lower expected wages, effectively trading match quality for employment security. These adjustments, along with the associated changes in assets and consumption among workers aged 20–40, are summarized in Table 6.

Table 6: Model Aggregates With and Without the Move Home Option

| **Variable** | **Baseline** | **No Co-Residence** | **Δ** |
|---|---|---|---|
| **Panel A: Percentage Point Changes** | | | |
| Employment Share | 0.9540 | 0.9556 | +0.0016 |
| Average Job-Finding Rate | 0.3926 | 0.4084 | +0.0158 |
| Tax Rate | 0.0651 | 0.0649 | –0.0002 |
| **Panel B: Relative Percentage Changes** | | | |
| Average Assets | 0.8784 | 0.8722 | –0.0071 |
| Average Consumption | 1.9366 | 1.9337 | –0.0150 |
| Average Housing Expenditure | 0.1148 | 0.1176 | +0.0244 |

**Note:** Computed among workers aged 20–40 to control for the ability to move home. Panel A measures changes between the baseline and no co-residence models in terms of absolute percentage point changes; Panel B measure these changes in relative percentage deviations. Restricting the move home option results in higher employment and job-finding rates, as well as lower taxes, assets, and consumption.

Constructing an environment in which all individuals are mechanically required to purchase housing raises employment primarily through higher job-finding rates. This counterfactual does not involve recalibrating the model or solving for a new equilibrium with different structural parameters, rather it holds the baseline economy fixed and removes the co-residence option, forcing everyone to behave as if independent living were the only feasible choice. Although this constraint increases employment, it does not generate meaningful gains in asset accumulation or consumption. The additional earnings associated with higher employment are largely absorbed by the mandatory housing expenditure, leaving households with little scope to build savings or improve their consumption.

The most notable quantitative result is the approximately 1.5 percentage point increase in the job-finding rate. In the model, submarket choice and savings behaviour are important margins through which workers manage unemployment risk and smooth consumption, but co-residence itself is a central insurance mechanism. Eliminating co-residence compresses the set of submarkets that are desirable to workers, since some submarkets that would otherwise be chosen become unattractive when housing costs are unavoidable. Without the ability to live at home, consumption while unemployed falls sharply, which raises the relative cost of unemployment and makes any level of unemployment risk more costly. As a result, workers shift toward submarkets with higher matching probabilities and lower piece-rates, reducing expected unemployment durations. The implied increase in the job-finding rate corresponds to 0.0989 model periods[3] (years), or about 5.1 weeks, of additional search before a match in the baseline economy relative to one where everyone must purchase housing. This model result mirrors empirical evidence from the NLSY97, which also shows longer search durations among those with access to co-residence.

Taken together, these results demonstrate that co-residence meaningfully reshapes job search behaviour by relaxing liquidity constraints and enabling workers to pursue higher-wage jobs. Removing this option compresses search durations, raises employment, and lowers wages, ultimately reducing consumption and wealth. This counterfactual highlights the central mechanism of the model: co-residence functions as a non-market insurance channel that allows workers to trade off longer unemployment spells for better long-run outcomes.

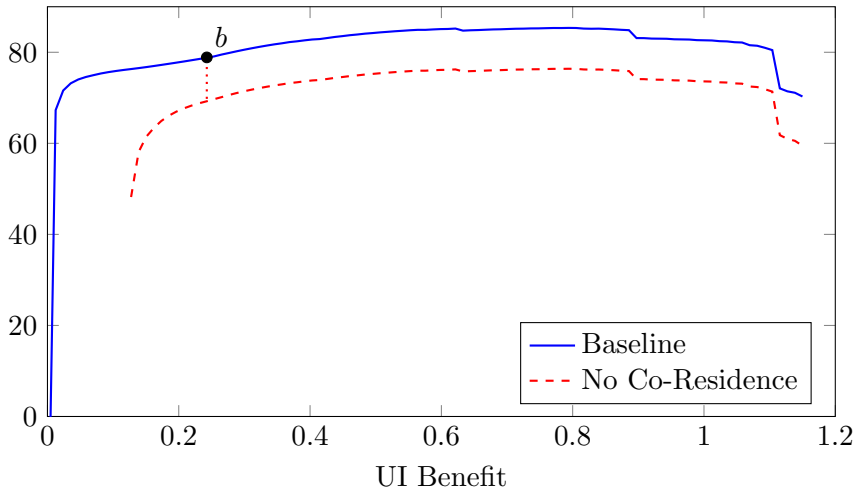
### 5.3 Welfare from Co-Residence

Beyond establishing the existence of co-residence as labour market insurance, it is worth examining how much this insurance channel is worth to consumers. The quantitative model is able to provide some insight using a counterfactual where, instead of having the option to move home, consumers receive an elevated unemployment insurance benefit such that they are indifferent to the availability

[3]Search durations are computed as 1/Job-Finding Rate. Baseline model: $1/0.39255 = 2.547$ years; no co-residence model: $1/0.40841 = 2.4485$ years. Difference: 0.0984 years or 5.145 weeks.

of parental co-residence.

Figure 4: Lifetime Expected Utility Conditional on UI Benefit




**Note:** The welfare associated with the move home benefit is largest when UI benefits are small. Note that no co-residence case is not well-defined for $b \leq \kappa_h$.

The vertical distance between the two plotted curves in Figure 4 is an estimate of welfare from the co-residence option. To evaluate this welfare in the context of an existing UI benefit, I measure the vertical distance ($\sim$9 utils) in terms of how much of a reduction in UI would induce the same utility reduction in an environment without co-residence. I find that, ex ante, workers are indifferent between the baseline economy with access to co-residence and an economy where co-residence is unavailable but UI benefits are raised by 71 percent. In other words, the option to move home is equivalent in welfare terms to a 71 percent increase in UI benefits.

The relative welfare benefit of the move home option is largest at low levels of UI. For example, the vertical distance between the two curves is much larger below $b$, the calibrated UI benefit, than above. This relationship implies that unemployment insurance and the move home option serve as alternate insurance mechanisms against labour market risk. In an environment with a generous UI system, non-market insurance through shared housing is relatively less important for the welfare of

workers. In contrast, when unemployment is more punishing through reduced benefits, the welfare implications of moving home are significant.

### 5.4 Welfare Across the Life Cycle

An under-examined aspect of parent-child co-residence is how the relationship between these two agents evolves as they both age. In Section 2, I presented rising parent needs later in life to motivate restricting co-residence in the model to adult children aged 40 and under. Even with this restriction designed to isolate labour market motives for co-residence, there is significant variation in move home flows by age. However, as shown in Table 7, this variation is not reflected in the value of the move home option across age groups. Older workers aged 36-40 value the option to co-reside almost as much as those aged 20-25, but do not exercise it nearly as often as younger workers. This is also true of workers in the middle of the age distribution: those aged 26-30 and 31-35 both value the move home option more than 20-25.

Table 7: Welfare by Age in Terms of UI Benefits

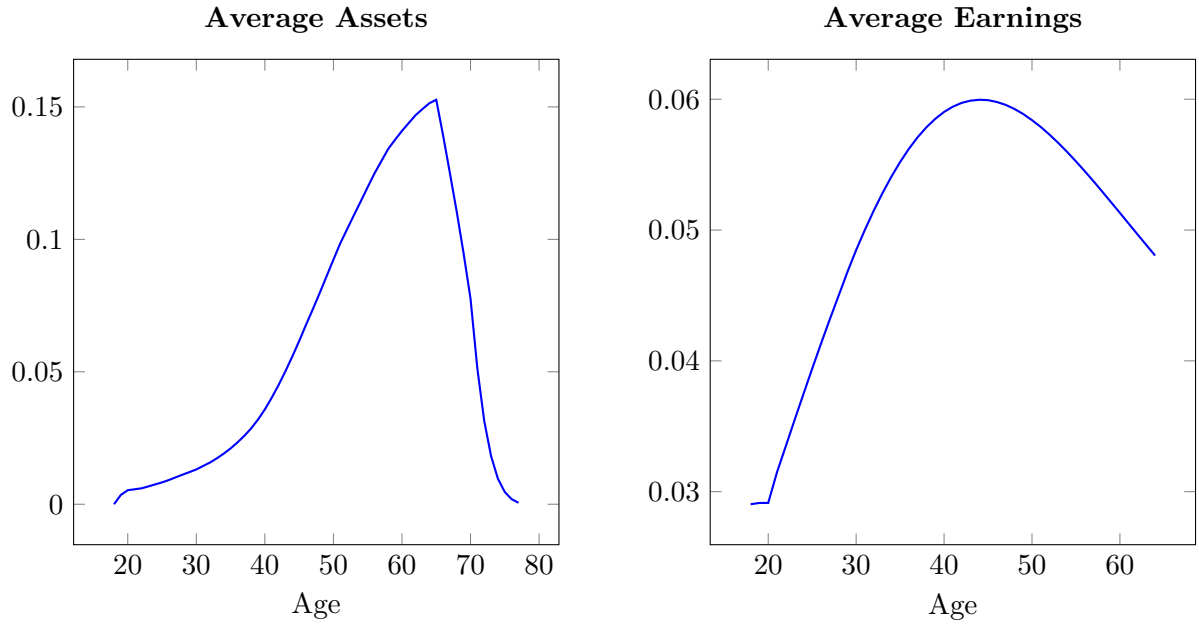
| **Age** | **Equivalent UI Benefit** | **Relative $\Delta$ (%)** |
|---|---|---|
| 20-25 | 0.13383 | 83.76 |
| 26-30 | 0.12741 | 93.02 |
| 31-35 | 0.12613 | 95.00 |
| 36-40 | 0.13768 | 78.62 |

**Note:** Column 2 reports the level of UI benefits that would leave each age group indifferent between the baseline economy, where individuals can move home, and an economy without the move-home option. These values show how much UI could be reduced, relative to the baseline UI benefit of 0.2459, while still delivering the same welfare as in the baseline model with co-residence. Column 3 reports the implied value of co-residence in the opposite direction, showing the percentage increase in UI benefits that would be required to make individuals indifferent between a lower-UI economy with co-residence and a higher-UI economy without co-residence. Combined, these two columns summarize the compensating variation in UI benefits associated with access to co-residence, expressed as two different but linked welfare comparisons.

One reason that the move home option is exercised relatively less as workers age is because they become better able to insure without co-residence. As workers age, they are on average better off in the current period. This is for two reasons: first, as shown in Figure 5, they accumulate assets

each period in which they work, and only begin to dissave when entering retirement. Second, wages are determined in part by individual-specific productivity which exhibits a hump-shaped trend over the life cycle.

Figure 5: Assets and Earnings by Age



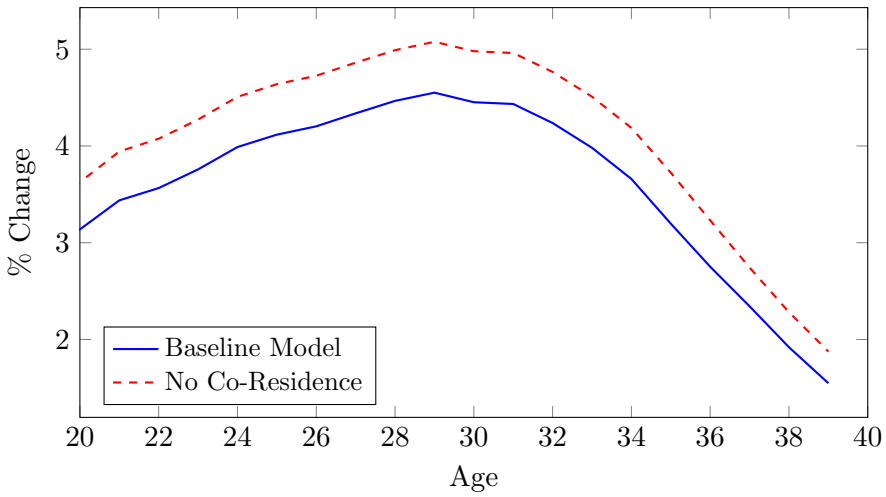

**Note:** Age profile of assets and earnings at the mean. Assets accumulate throughout the worker's life until they choose to dissave rapidly in retirement. Earnings are computed as after-tax labour income among the employed. Earnings rise early in life but decline as workers approach retirement for two reasons. First, the age component of productivity is sharply increasing at young ages before leveling off in late-middle age and declining slightly before retirement. Second, wages are partially determined via a posting decision by firms. Since age is both observable and relevant for future earnings flows, firms will naturally offer higher wages (piece-rates) to younger workers in expectation of output in future periods in the absence of a job destruction shock.

However, workers are not necessarily better off in lifetime terms as they age. Since this model environment features a finite horizon life-cycle, the present expected utility of an individual is declining as they age and approach mortality. Additionally, in the absence of savings, retirement will result in a large consumption loss relative to working periods because public pensions are smaller than employment income, but larger than unemployment benefits. The potential negative effects of retirement on consumption create an environment where aging is ambiguous in welfare

terms. Older agents can save more and have higher wages, however, they have fewer additional years of work where they can accumulate assets and enjoy those heightened wages. This results in an environment where older workers are relatively wealthier but also more vulnerable to future income losses.

These dynamics are reflected in the consumption responses to job separation. Figure 6 shows that consumption losses on an EU transition decline with age, because older workers rely more heavily on accumulated assets to smooth consumption. Yet the option to move home reduces consumption losses at all ages, indicating that co-residence provides meaningful insurance even for workers with substantial savings.

Figure 6: Consumption Losses on Job Separation




**Note:** One way to examine the value of an insurance mechanism is to evaluate behaviour following a negative income shock (unemployment). In the first panel, high consumption losses when young are driven by a desire for asset accumulation. As workers age and become wealthier, they increasingly substitute assets to offset potential declines in consumption. When controlling for difference in income, it is clear that consumption losses on an unemployment shock are monotonically declining beyond age 22. However, there are still insurance benefits to co-residence at all ages. This can be seen through the vertical distance between consumption losses in the baseline model and in the counterfactual with no option to co-reside.

When I restrict the analysis to workers who move home following an EU transition in the baseline

Table 8: Consumption Change on EU Transition

| **Age** | **Baseline Model (%Δ)** | **No Co-Residence (%Δ)** |
|---|---|---|
| 31 | +14.9 | -14.9 |
| 32 | +8.9 | -19.4 |
| 33 | +5.4 | -22.0 |
| 34 | +3.4 | -23.5 |
| 35 | 0 | -26.0 |
| 36 | -3.6 | -28.7 |
| 37 | -5.5 | -30.1 |

**Note:** Consumption losses conditional on an EU transition in the baseline and no co-residence model environments. At some ages ($< 35$), consumption rises as workers age regardless of employment status. This is due to positive expectations of future wages and differences in future opportunities to work based on time endowments. For older workers, a job separation is associated with consumption losses relative to the previous period. However, in an environment without the co-residence option all workers reduce consumption during an EU transition. At ages where consumption falls in both the baseline and counterfactual scenarios, losses are larger in the economy without the option to move home.

model, meaning they live independently when employed and return home when unemployed, I find additional evidence that the option to move home is used as a consumption-smoothing mechanism. Table 8 reports consumption changes for this selected group under two environments: the baseline model, in which they are able to move home, and a counterfactual in which they are forced to live independently. This comparison is constructed by identifying individuals who choose to move home after an EU transition in the baseline economy and then tracking those same individuals in the counterfactual where co-residence is not available. In the baseline model, an EU transition leads to a consumption loss for older workers, whereas younger workers experience a slight increase due to their stronger expectations of future high-wage employment. Across all ages, however, consumption while unemployed is higher in the baseline model than in the counterfactual. This indicates that the option to move home provides meaningful consumption insurance not only in the aggregate, but specifically for the children who intend to use co-residence following an EU transition.

The welfare value of co-residence is both quantitatively large and persistent across the life cycle. Older workers exercise the option less frequently, but the option value is still beneficial. Co-residence therefore remains an important form of non-market insurance well into middle age.

# 6 Conclusion

This paper provides new evidence that co-residence between parents and adult children operates as a meaningful form of non-market insurance well into middle age. Using linked panel data from the Health and Retirement Study, I document that workers up to age 40 exhibit a substantially higher likelihood of returning to the parental home following a job separation, even after conditioning on a wide range of parental characteristics that become increasingly salient with age. These include parental health diagnoses, functional limitations, medical histories, and the provision of eldercare by specific children. The persistence of co-residence flows across the life cycle, and their robustness to controls for parental need, indicates that shared housing remains an important response to labour market shocks long after the early career period typically emphasized in the existing literature.

To quantify the value of the option to move home, I develop a structural life-cycle model of directed search in which workers choose both a submarket in which to search and a living arrangement, either independently or with a parent. The model embeds a fixed cost of independent housing and allows workers to avoid this cost by co-residing with parents until age 40, at the expense of a utility loss from reduced independence. This framework captures the central trade-off facing workers: co-residence relaxes liquidity constraints and increases their ability to search for higher-wage jobs, but it also entails non-pecuniary costs. The calibrated model successfully reproduces the empirical co-residence flows observed in the HRS, including the elevated propensity to move home following an employment-to-unemployment transition and the decline in these flows with age. The model also clarifies the mechanism behind these patterns: co-residence allows workers to tolerate longer unemployment spells, search more selectively, and accept higher wages.

Counterfactual exercises highlight the quantitative importance of this mechanism. Removing the option to move home shortens unemployment durations by approximately eight weeks and increases job-finding rates, but at the cost of lower accepted wages, reduced consumption, and diminished asset accumulation. These results underscore that co-residence affects not only the level of unemployment but also the composition of matches formed in equilibrium. The welfare

consequences are substantial. For an ex ante worker, access to co-residence is equivalent to a 71 percent increase in unemployment insurance benefits. Moreover, the welfare value of co-residence remains large for workers aged 36–40, who treat the option as equivalent to nearly an 80 percent increase in UI benefits. This pattern reflects the interaction between rising asset holdings, declining future earnings horizons, and the heightened vulnerability of older workers to income shocks.

Several avenues for future research emerge from this analysis. Empirically, there is value in investigating how the prevalence of this co-residence insurance mechanism varies across local housing markets. This paper relies on an abstraction of a common housing cost experienced by all workers: relaxing that assumption or imposing variation in housing costs may provide further insight. Quantitatively, this analysis could be extended to account for strategic interactions between parents and children to examine issues of under-saving and differential support to siblings. Further, allowing for two-sided bargaining in wage formation would allow for measurement of match quality on the firm side instead of limiting the examination to wages on the worker side. For policymakers, understanding how informal non-market insurance interacts with public insurance programs remains an outstanding question. This analysis demonstrates that co-residence is a quantitatively significant component of the household's insurance portfolio and future work can build on this foundation to explore its broader macroeconomic and distributional implications.

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# A Appendix

## A.1 HRS: Pr(Move Home) Probit Estimation

Table 9 reports the full probit introduced in Section 2.2. Individual observations correspond to pairs of parents and children. Standard errors are clustered at the pair-level and the McFadden's $R^2$ is 0.1307. A joint Wald test of significance on all coefficients rejects the null hypothesis with extreme confidence. The formal specification is:

$$Pr(MoveHome|\boldsymbol{X}) = \Phi(\alpha_0 + \boldsymbol{\alpha}\ \boldsymbol{Year} + \boldsymbol{\beta}\ \boldsymbol{X^c} + \boldsymbol{\gamma}\ \boldsymbol{X^p} + \boldsymbol{\delta}\ \boldsymbol{X^h} + \boldsymbol{\eta}\ \boldsymbol{X^e})$$

The vectors of characteristics ($\boldsymbol{X}$) are as follows:

- Child characteristics: $\boldsymbol{X^c}$ = {Income, Education, Age, Gender, Employment Transition, Marital Status, Parental Status}
- Parent characteristics: $\boldsymbol{X^p}$ = {Income, Assets, Primary Residence Value, Education, Employment}
- Parent health characteristics: $\boldsymbol{X^h}$ = {High Blood Pressure, Diabetes, Lung Disease, Heart Disease, Ever Had Stroke, Ever Had Cancer}
- Eldercare characteristics: $\boldsymbol{X^e}$ = {Child Helps with ADLs, Child Helps with IADLs, Child Intends to Help in Future}

Table 9: Pr(Move Home) Probit: Full Estimation

| **Independent Variable** | **AME** | **p-value** |
|---|---|---|
| **Year** | | |
| 2000 | *Base* | |
| 2002 | 0.00246 | 0.182 |
| 2004 | 0.00483 | 0.023 |

*Continued on next page*

Table 9: Average Marginal Effects on Pr(Move Home) (continued)

| **Independent Variable** | **AME** | **p-value** |
|---|---|---|
| 2006 | 0.00024 | 0.898 |
| 2008 | 0.00738 | 0.001 |
| 2010 | 0.00369 | 0.076 |
| 2012 | 0.00466 | 0.016 |
| 2014 | 0.00504 | 0.020 |
| 2016 | 0.00782 | 0.000 |
| 2018 | 0.01095 | 0.000 |
| **Child Income** | | |
| <10K | *Base* | |
| 10–35K | −0.01155 | 0.000 |
| 35–70K | −0.02003 | 0.000 |
| 70–100K | −0.02035 | 0.000 |
| 100K+ | −0.02411 | 0.000 |
| **Child Education** | | |
| Less Than High School | *Base* | |
| High School Graduate | −0.00433 | 0.193 |
| Partial College Completion | 0.00203 | 0.563 |
| College and Above | −0.00141 | 0.669 |
| **Child Age** | | |
| 20–25 | *Base* | |
| 26–30 | −0.00629 | 0.001 |
| 31–35 | −0.00837 | 0.000 |
| 36–40 | −0.01244 | 0.000 |
| **Child Employment Transition** | | |
| E–E | *Base* | |
| E–U | 0.01588 | 0.000 |
| U–E | 0.00353 | 0.040 |
| U–U | 0.00331 | 0.079 |
| Child Gender (Female) | −0.00338 | 0.000 |
| Child Marital Status | −0.02618 | 0.000 |
| Child Parental Status | 0.00034 | 0.790 |
| Child Helps w/ ADLs | 0.01627 | 0.007 |
| Child Helps w/ IADLs | 0.01269 | 0.007 |
| Child Intends to Help in Future | 0.01125 | 0.000 |
| Parent Income | $6.22 \times 10^{-10}$ | 0.802 |

*Continued on next page*

Table 9: Average Marginal Effects on Pr(Move Home) (continued)

| **Independent Variable** | **AME** | **p-value** |
|---|---|---|
| Parent Assets | $-1.84 \times 10^{-9}$ | 0.021 |
| Parent Home Value | $3.86 \times 10^{-9}$ | 0.002 |
| **Parent Education** | | |
| Less Than High School | *Base* | |
| GED | −0.00300 | 0.263 |
| High School Graduate | 0.00412 | 0.028 |
| Partial College Completion | 0.00276 | 0.135 |
| College and Above | 0.00397 | 0.037 |
| **Parent Employment** | | |
| Not Working | *Base* | |
| Part-time Work | 0.00160 | 0.457 |
| Full-time Work | 0.00318 | 0.008 |
| Parent has High Blood Pressure | −0.00091 | 0.435 |
| Parent has Diabetes | −0.00007 | 0.960 |
| Parent has ever had Cancer | −0.00107 | 0.464 |
| Parent has Lung Disease | 0.00067 | 0.689 |
| Parent has Heart Disease | 0.00069 | 0.583 |
| Parent has ever had a Stroke | 0.00518 | 0.005 |

**Note:** Average marginal effects are expressed in relative deviations from the mean share of children who move home (0.007518). McFadden's $R^2$: 0.1307; Wald $\chi^2$: 1236.79

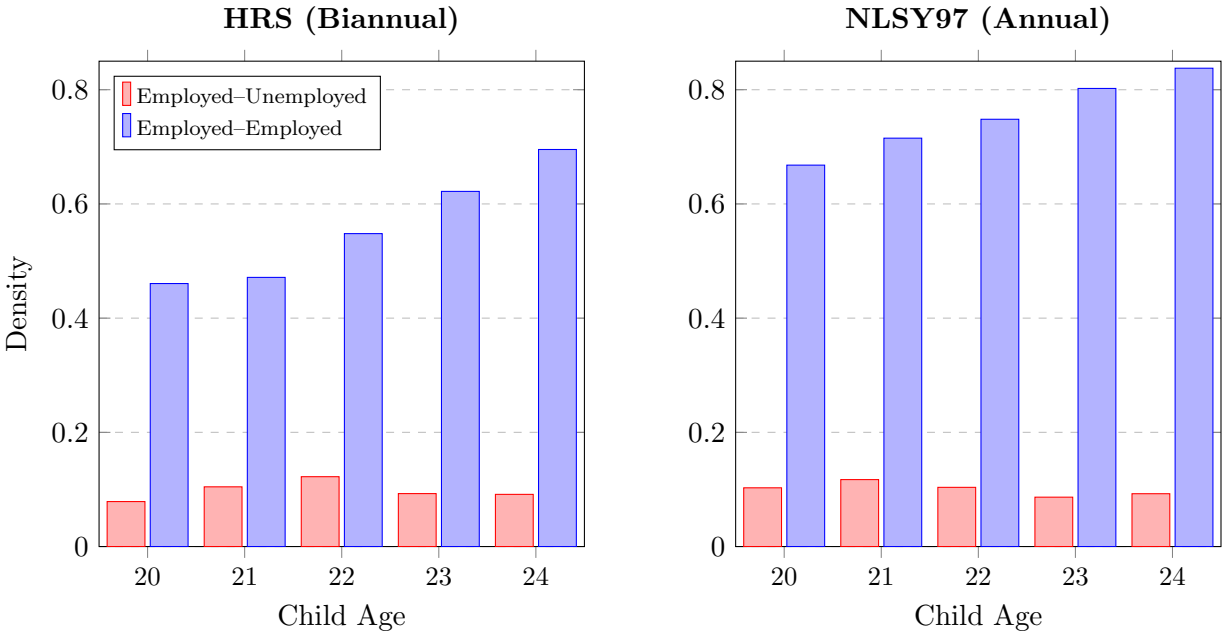
## A.2 Comparability of the HRS and NLSY97

Although the HRS and NLSY97 differ in design and target populations, the key moments relevant for this analysis are highly comparable. Both datasets report co-residence status and employment transitions, and both show a pronounced increase in the likelihood of moving home following an Employed-to-Unemployed (EU) transition. The age profiles of move-home rates also align closely across the two datasets, with the probability of returning home declining steadily from the early twenties to age 40. These similarities indicate that the labour-market-driven co-residence patterns observed in the HRS reflect a broader empirical trend.

To evaluate the comparability of the two main datasets I use in this analysis, I present several key

moments. Figure 7 plots the relative mass of workers moving home conditional on job separation (EU) or job retention (EE). In both datasets, EU workers are significantly more likely to move home than EE workers. This strengthens the argument for co-residence as labour market insurance, because it shows the main empirical observation in the HRS is also present in the more commonly used NLSY97.

Figure 7: Comparison of Employment Transitions Between HRS and NLSY97 (2004)

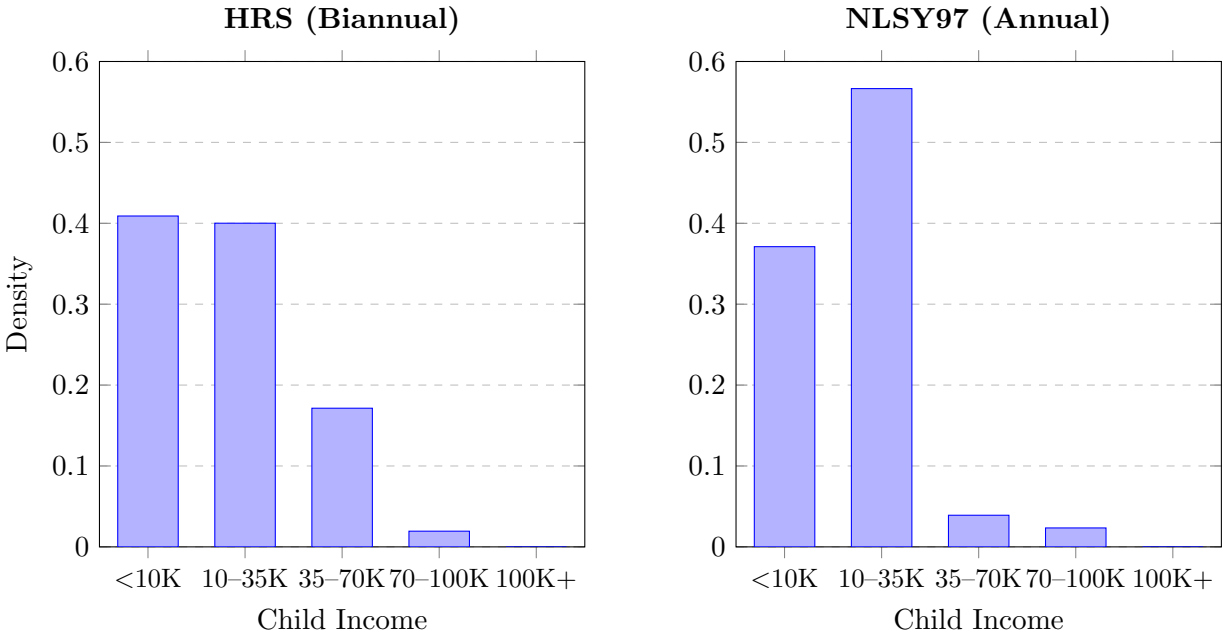



**Note:** As workers age, the relative mass of EE increases alongside job tenure. Estimates for EE densities vary significantly as a result of differences in survey composition and methodology.

Figure 2 of Section 2.4 plots the relative density of workers conditional on job transition. In both datasets, job tenure and retention increase as workers gain valuable experience. Similar to the previous plot, there is a distinct difference in levels between the two datasets. One possible explanation for this variation is differences in the frequency of the datasets. The HRS is limited to reporting once every two years, while the NLSY97 is primarily annual and in some cases monthly or weekly. Transforming the NLSY97 to match the frequency of the HRS reduces these differences in levels to some degree but inconsistencies remain.

Another explanation is variation in the sample composition between the two surveys. For example, the HRS sample is targeted to be representative of Americans over 50, while the NLSY97 is a representative sample of individuals born between 1980 and 1984. These surveys target very different populations of interest so it is unsurprising that some moments, notably employment flows, are very different. It is encouraging then that even with these differences in sample composition that two facts remain: first, there is suggestive evidence that workers use the option to move home as labour market insurance. Second, employment flows, at least with respect to increasing job tenure across the life-cycle, are consistent across both the HRS and NLSY97.

Figure 8: Comparison of Income Among 20–25 Year-Olds in the HRS and NLSY97 (2004)




**Note:** Income distributions among adult children in the HRS and NLSY97. The HRS reports child income as a categorical variable and I convert the NLSY's continuous income to align the two datasets.

Income distributions from the two datasets share similar features. Figure 8 shows the income distribution among 20-25 year-olds in the year 2004. Incomes are binned to align with the categorization present in the HRS. In both datasets, very few workers report incomes above $\$70,000$ and virtually none earn above $\$100,000$. Respondents in the HRS are slightly richer, reflected in a

larger density in the middle ($\$35 - \$70K$) bin relative to the second-to-lowest ($\$10 - 35K$) category. Individuals with the lowest income are slightly more common in the HRS, but this difference is sufficiently small as to not be economically meaningful.

### A.2.1 HRS versus Biannual NLSY97

One limitation to comparing moments from the HRS and NLSY97 is the differences in frequency between the two surveys. This is relatively straightforward to correct simply by transforming the NLSY into a biannual dataset. This process is akin dropping approximately half of the observations from the NLSY97 and is thus very costly from the perspective of statistical precision. Nonetheless, this transformation can be valuable to understand the role of survey frequency in differences in moments between the two sources. Figure 9 shows the share of workers who move home conditional on EU and EE transitions. In the previous, I pointed out a difference in levels between the two datasets. Adjusting the NLSY97 to a biannual frequency exacerbates these differences in levels, resulting in move home shares in the NLSY97 that are significantly larger than when treating the data annually. Figure 10 reports employment flows under a similar adjustment.

## A.3 NLSY97: Search Duration and Co-Residence

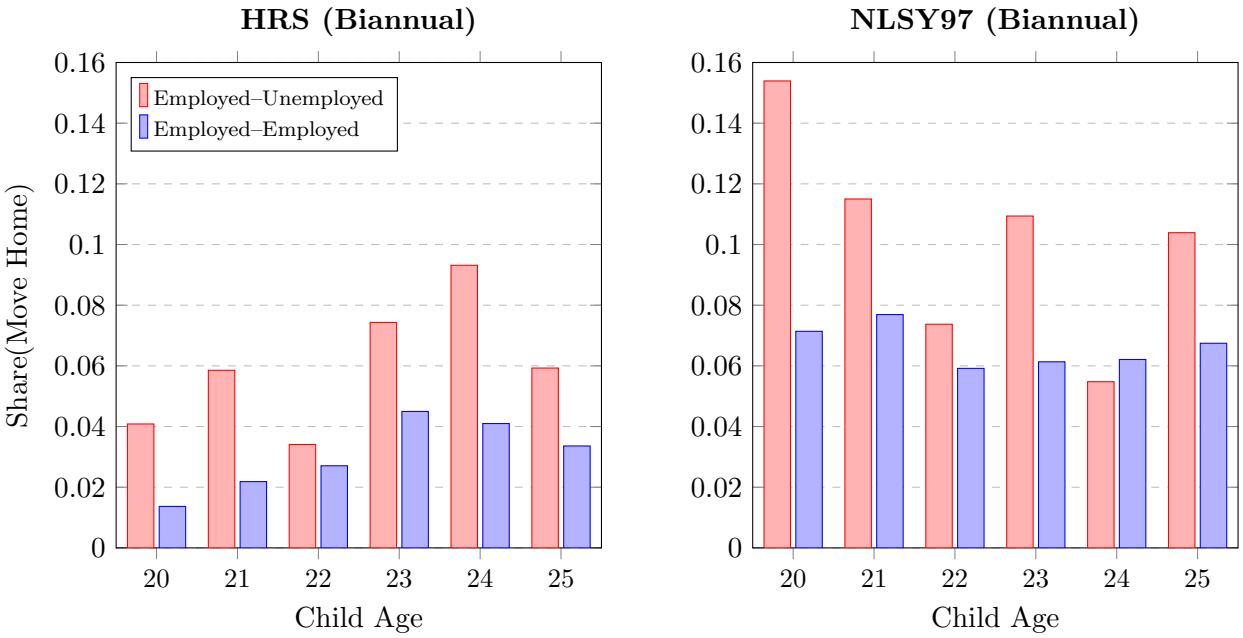
Table 10 reports the full OLS specification first presented in Section 2.4. I estimate a regression equation of the form:

$$\begin{aligned} SearchDuration = \beta_0 + \beta_1 Co-Residence + \beta_2 PastIncome + \boldsymbol{\beta_3 Year} + \boldsymbol{\beta_4 Age} \\ + \boldsymbol{\beta_5 Education} + \beta_6 Gender + \beta_7 Transfer from Parent + \epsilon \end{aligned}$$

Where **Year**, **Age**, and **Education** are vectors of indicator variables corresponding to $\boldsymbol{\beta_2}$, $\boldsymbol{\beta_3}$, and $\boldsymbol{\beta_4}$ vectors of coefficients for those indicators, respectively.

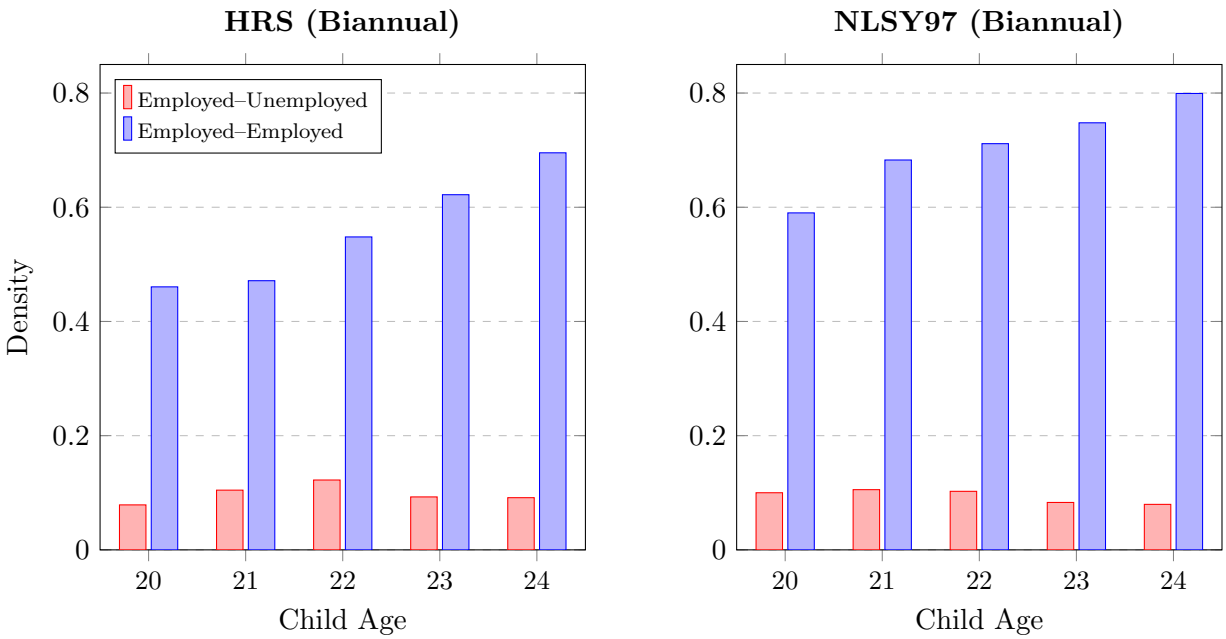
Age and year fixed-effects are generally not statistically significant with one exception: 2009.

Figure 9: Comparison of Move Home Trends Among the HRS and NLSY97




**Note:** For most of their early twenties, EU workers are more likely to move home relative to their EE counterparts. Adjusting the NLSY97 to mimic the biannual nature of the HRS does not shrink the differences in levels between the two datasets, but instead increases them.

Figure 10: Comparison of Employment Transitions Between HRS and NLSY97 (2004)




**Note:** Biannually transforming the NLSY97 reduces the difference in levels slightly, but comparability issues persist.

This is not entirely surprising, as workers at this time were experiencing the effects of the Great Financial Crisis, and as during any recession the most severe employment effects are disproportionately borne by low-productivity and low-tenure workers. These estimates are relatively imprecise, but the marginal effect of living at home on search duration are significant and economically meaningful. Additional measures of overall model fit such as the coefficient of determination ($R^2$) and a joint test of coefficient significance ($F$-test) indicate the model captures statistically significant relationships, even if the overall explanatory power is limited.

Table 10: OLS Estimation of Co-Residence and Search Duration: Full Results

| **Independent Variable** | **Coefficient** | **p-value** |
|---|---|---|
| Co-Residence | 0.762 | 0.016 |
| Past Income | 0.0000 | 0.009 |
| **Year** | | |
| 1999 | *Base* | |
| 2000 | −0.142 | 0.850 |
| 2001 | 0.755 | 0.337 |
| 2002 | 0.830 | 0.275 |
| 2004 | 1.543 | 0.075 |
| 2005 | 0.227 | 0.795 |
| 2006 | 0.139 | 0.879 |
| 2007 | 0.268 | 0.782 |
| 2008 | 1.358 | 0.197 |
| 2009 | 3.886 | 0.001 |
| **Age** | | |
| 18 | *Base* | |
| 19 | 0.529 | 0.743 |
| 20 | 0.667 | 0.683 |
| 21 | 0.762 | 0.640 |
| 22 | 1.179 | 0.480 |
| 23 | 1.549 | 0.360 |
| 24 | 2.302 | 0.180 |
| 25 | 3.476 | 0.049 |
| 26 | 2.353 | 0.192 |
| 27 | 3.925 | 0.038 |
| 28 | 1.423 | 0.468 |
| 29 | 5.640 | 0.031 |
| **Education** | | |
| Less Than High School | *Base* | |
| HS Graduate | −0.681 | 0.134 |
| Some College | −0.987 | 0.238 |
| Post-Secondary and Above | −1.149 | 0.043 |
| Gender (Female) | −0.696 | 0.008 |
| Enrolled in School | −1.834 | 0.000 |
| Financial Transfer from Parent | 0.022 | 0.947 |

**Note:** $R^2$: 0.0618, $F$-statistic: 7.14